

tv: cinema or home theater or theater

utilityroom/toolroom: utility room or tool room

workout/gym/exercise: gym

F Additional Grounded Navigation Results

The performance of the LLM and Tag Map grounded navigation pipeline is evaluated on a set of user queries that could be asked of a home assistant robot. The evaluation is done on a scan of the lab environment used for the real robot experiments in Sec.5.4. We assume that the environment is entirely traversable such that the robot can reach any goal location from any starting location.

A total of 25 user queries were evaluated. The grounded navigation for a query is considered successful if the LLM grounded on the Tag Map context suggests an instance of a semantic class that is relevant to addressing the query and if that instance is correctly localized from the Tag Map. The Tag Map grounded navigation was successful in 21 of the 25 test queries. The detailed results for all test queries are reported in Tables 9, 10, and 11.

Query	Suggested Entity	Successful?	Example Goal Viewpoint
Please heat up my lunch	microwave	✓	
Please refill my water bottle	faucet	✓	
Please turn on the lights	light switch	✓	
Please put away these dirty dishes	dish washer	✓	
Can you bring out the ice cream It's the other fridge, not the main one	fridge	✓	

The LLM localizes two fridges and a kitchen region from the tag map. It selects the fridge further from the kitchen as it associates the fridge in the kitchen as the main fridge.			
Ouch, I got a paper cut	soap	✓	
Can you go check if there's any mail for me?	mailbox	✗	

The image tagging model mistakenly recognizes a fire alarm as a mailbox.			
Get me something hot to drink	coffee machine	✓	

Table 9: Tag Map grounded navigation results for the test user queries (1/3).

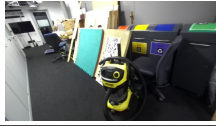
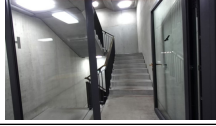
Query	Suggested Entity	Successful?	Example Goal Viewpoint
I spilled some water on the floor	paper towel	✓	
I spilled some sand on the floor	vacuum	✓	
Prepare somewhere for me to take a nap	couch	✓	
Can you go downstairs to meet the pizza delivery driver?	stairwell	✓	
It's raining outside, can you bring me something	umbrella	✗	
The image tagging model mistakenly recognizes an umbrella in the image.			
It's getting quite cold in here	boiler	✗	
The boiler tag describes a kettle, but the LLM misinterprets it as a home heating boiler.			
It's getting quite cold in here, could you bring me something?	coat	✓	
In contrast to the previous query, the LLM interprets this query as bringing an item to warm the user instead of trying to warm the space			
Take my phone and charge it	charger	✓	
Help! my shirt is on fire	extinguisher	✓	
Please prepare a location for the party this evening	terrace	✓	

Table 10: Tag Map grounded navigation results for the test user queries (2/3).